

Bier's Block in Children's Emergency

Bier's Block is a form of regional anesthesia of the forearm and/or the leg.

Background

Bier's block is an effective, time-proven, safe form of regional anesthesia. It was first introduced by August Bier in 1908. Its popularity waned but was revived by Holmes in 1963.

The premise of its use is exsanguination of the limb (by gravity) followed by arterial tourniquet which isolates the limb from the circulation. The limb is then anesthetized by injection of local anesthetic to the venous system.

Usage (limited to upper limb in our practice)

1. Manipulation and reduction of forearm fractures
2. Elbow dislocations
3. Lacerations of forearm (multiple / complicated)
4. Expected duration of procedure < 45 minutes

Equipment

1. Intravenous cannulation set
2. Double tourniquet with pneumatic cuff
3. Triangular bandage
4. Venous cannula
5. Resuscitation equipment
6. Cardiac and pulse oximetry monitoring
7. Requisites for backslab application

Contraindications

1. Lignocaine allergy – for e.g. if any dental procedures under local anesthesia have been performed before and if any reaction during such procedures.
2. Weight < 25kg and Age < 8 years
3. History of cardiac disease (arrhythmias)
4. History of epilepsy/seizures (relative)
5. Fractures where pulse needs to be monitored as a guide to reduction (e.g. severe supracondylar fracture where brachial artery is at risk)
6. Severe hypertension where leakage of tourniquet is a risk (rare in paediatrics)
7. Severe peripheral vascular disease (rare in paediatrics)
8. Sick Cell Disease (rare in paediatrics)
9. Diseases over the procedural limb – lymphedema, infection
10. Methemoglobinemia
11. Uncooperative patient

Procedure

1. Always ensure senior supervision if this is the first time performing the Bier's block
2. Always explain to the patient and parents what to expect before, during and after the procedure.
3. 2 assistants are always needed – one to apply the pneumatic cuff and monitor vital signs and another to assist in the M&R. If the procedure is for lacerations then at least 1 assistant is needed.
4. Ensure resuscitation drugs are present – namely normal saline for fluid bolus, lorazepam for seizures and intralipid (Intralipid may be obtained from Anaesthesia Dept).
5. Calculate dose required of **0.5% lignocaine** (pls note: lignocaine in CE comes as a 1% preparation) as follows:
 - a. Wt > 25kg
 - b. Dose of 0.5% lignocaine = 0.4ml/kg
 - i. **This is equivalent to 2mg/kg, to a maximum of 100mg absolute.**
 - c. Final total volume MUST NOT EXCEED 20ML (i.e. Once patient's weight is > 50kg, max volume of 0.5% lignocaine of 20ml is used)
 - d. **Toxic dose of lignocaine is 3mg/kg**
6. Prepare 20ml of 0.5% lignocaine by drawing out 10ml of 1% lignocaine and mixing with 10ml of normal saline **7. Discard** the volume that is not needed from the resultant 20ml of 0.5% lignocaine
8. Double check dose with another colleague before administering dose

Example 1:

Wt 30kg

Volume of 0.5% lignocaine = $0.4 \times 30 = 12\text{ml}$

20ml of 0.5% lignocaine prepared; **Discard 8ml**. i.e. $20\text{ml} - 8\text{ml} = 12\text{ml}$ remaining (desired volume)

Check the dose with assistant before administration

Example 2:

Wt 60kg

Volume of 0.5% lignocaine = $0.4 \times 60 = 24\text{ml}$ BUT maximum allowed is 20ml

Total volume administered = mixture of 10ml (1% lignocaine) + 10ml (normal saline) = 20ml (0.5% lignocaine)

9. Set ONE venous cannula on ipsilateral side of limb manipulation. There is no need to set a second cannula.
10. Check radial pulse and capillary refill of injured arm to document presence of radial pulse and good capillary refill
11. Exsanguinate the limb – raise above the level of the heart for 1 minute

12. Check vitals of patient, notably blood pressure and attached ECG leads to monitor cardiac rhythm.
13. Apply pneumatic cuff to patient's arm to at least 15mmHg **above** patient's systolic blood pressure. (Absolute value usually at least 225mmHg)
 - a. It is good practice to tie a tourniquet above the pneumatic cuff with a triangular bandage, in case there is pneumatic cuff failure/slippage during the procedure.
14. Palpate the radial pulse before inflating cuff and feel for disappearance of radial pulse as cuff is inflated
15. Slowly administer the 0.5% lignocaine over 60 seconds. Warn the patient there will skin discolouration/blanching spots.
- a. RECORD TIME OF INFUSION OF LIGNOCAINE
 - b. Tell the patient to notify you if they feel any peri-oral numbness, tingling, any lightheadedness or giddiness, tinnitus or blurred vision.
 - c. Maintain verbal contact with the patient – watching out for dysarthria or impaired consciousness.
 - d. These are all signs of lignocaine toxicity. If present, stop the lignocaine infusion immediately and recheck pneumatic cuff.
16. Wait at least 5 minutes before starting the procedure. Anesthesia of the infused limb typically sets in after 5 minutes.
17. If there is inadequate anesthesia, flush the venous cannula with another 10-15ml of normal saline.
18. Remove the venula after successful regional anesthesia and secure hemostasis. Proceed with procedure.a. Procedure should not extend beyond 30 minutes, after which the patient will usually have tourniquet pain
19. The tourniquet should be left on for a minimum of 20 minutes from the time of lignocaine infusion and a maximum of 45 minutes.
20. Upon completion of the procedure, deflate cuff completely.
21. Monitor the patient closely during period of cuff deflation.
22. Send the patient for post reduction X-ray only AFTER cuff is deflated
23. A second attempt at Bier's block is NOT allowed.
24. Observe in CE for an hour before discharge.

Symptoms/Signs of Lignocaine Toxicity

CNS (occurs at lower dose than CVS toxicity)

1. Early
 - a. Perioral paresthesia, metallic taste in mouth
 - b. Visual and auditory disturbance (unable to focus, tinnitus)
 - c. CNS agitation or confusion
2. Late
 - a. Seizures
 - b. Coma

CVS

1. Shortness of breath
2. Diaphoresis

3. Hypotension
4. Dysrhythmias (usually ventricular)

Hematological

1. Cyanosis - Methaemoglobinemia (rare)

Allergic

1. Rash
2. Urticaria
3. Anaphylaxis (rare)

Management of Severe Lignocaine toxicity (e.g. haemodynamic instability)

1. Stop lignocaine injection if still ongoing, confirm tourniquet inflation
2. Acute resuscitation
 - a. ABC
 - b. Oxygen
 - c. Call for help
3. Benzodiazepines for seizures
4. Lipid emulsion therapy after airway management completed
 - a. Bolus of 1.5ml/kg of lipid emulsion (E.g. Intralipid 20%), AND, start lipid emulsion infusion at 15ml/kg/hr
 - b. Review after 5min, if still hemodynamically unstable, repeat bolus of 1.5ml/kg AND double rate of lipid emulsion infusion
 - c. Review after 5min, if still hemodynamically unstable, repeat final bolus of 1.5ml/kg of lipid emulsion.
 - d. Maximum cumulative dose of lipid emulsion is 12ml/kg

Discharge criteria 1. Absence of:

- a. peri-oral numbness, tingling, any lightheadedness or giddiness, blurred vision, tinnitus, dysarthria or impaired consciousness
- b. Dysrhythmias
2. Normal age-specific vitals
3. Normal limb circulation

Special Precautions

1. Ensure a good venula is inserted – extravasation of lignocaine is severe and causes tissue necrosis.
2. Always ensure pneumatic cuff is fully inflated and one assistant to continually ensure the cuff has not slipped or deflated.

References:

1. NUH SOP: NUH-SOP-MCP-EMD-001
2. NUH SOP: NUH-SOP-NSG-EMD-040
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Author: Dr Lim Yang Chern (Nov
2015) Reviewed by: Dr Chew Su Yah
(Nov 2015)

Updated by: Dr Tan Ze Lei (May 2023)

